

UNDERSTANDING YOUR CPU

THIS IS NOT GOING TO BE EASY

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We are completely oblivious to the way the actual tasks we use a computer to perform get executed at the hardware level. We know that a good portion of the tasks speed up when you upgrade to a better CPU or when you get a graphics card to improve the visual fidelity in video games or when you add faster RAM or when you add an SSD. There are so many things one can do to get more performance from their computers but the CPU (Central Processing Unit) or the Processor is inarguably the biggest contributor in this aspect. And yet, the vast majority of us know so little about the humble processor. Let's change that.

We're not going into the details of how transistors work, that's an extra level of detail which you can go figure out for yourself. Or perhaps you can check out the jargon busters we've included in each article to get a better understanding. For now, let's look at the basic functionality of the processor.

THE PROCESSOR SUPREME

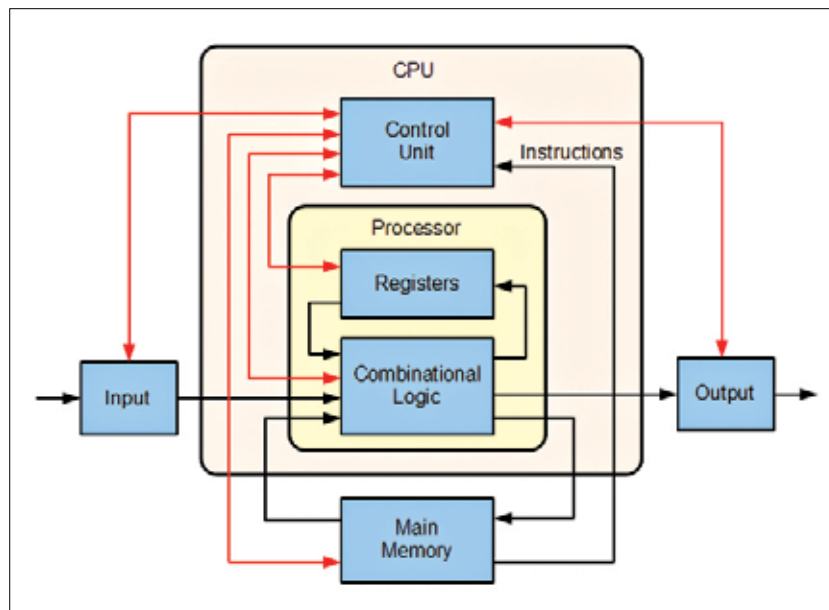
While the form factor, the ISA (Instruction Set Architecture) and many other aspects have changed over the years, the fundamental workings of the processor remains to be the same for both Intel and AMD. At the centre

of it all, the processor has the ALU (Arithmetic Logic Unit) and the (CU) Control Unit working on input data sourced from the memory. If you've ever come across a book with any mention about CPUs then the basic diagram should be all too familiar.

The Control Unit is like the manager for the CPU. It takes a look at all the data that has come in, gets the instruction from the user as to what must be done with the data, and then figures out a way to get the job done. While the data you input into a computer might be in the form of text, audio, video, or a combination of all, the processor only sees the data as very specific instructions. These instructions are arrived at after the input you provide goes

through multiple layers of processing before it arrives at the processor. This is because each processor is hardwired to understand basic instructions and everything has to be translated to match that. These basic instructions that are hardwired in the CPU are called the Instruction Sets, and every processor supports multiple instruction sets. You can learn more about these in the following pages. All you need to remember is that the Control Unit figures out when to execute instructions, how to execute the instructions and works out the sequence in which they are to be executed.

Then comes the ALU (Arithmetic Logic Unit) which is tasked with performing arithmetic and bitwise logic



Is this it? Seems simple.